

# PERIPHERAL NERVES

THE BODY'S NETWORK OF PERIPHERAL NERVES COMMUNICATES WITH THE BRAIN AND SPINAL CORD. SENSORY FIBRES CARRY MESSAGES FROM SENSE AND INTERNAL ORGANS, WHILE MOTOR FIBRES CONTROL MUSCLE AND GLAND ACTIVITY.

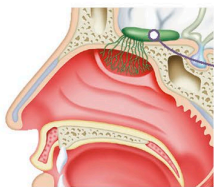
## CRANIAL NERVES

The 12 pairs of cranial nerves connect to the brain directly, not via the spinal cord. Some perform sensory functions for organs and tissues in the head and neck, while others have motor functions. The nerves with predominantly motor fibres also contain some sensory fibres that convey information to the brain about the amount

of stretch and tension in the muscles they serve, as part of the proprioceptive sense (see p.73). Most of the cranial nerves are named according to the body parts they serve, such as the optic nerves (eyes). By convention, the nerves are also identified by Roman numerals, so the trigeminal nerve, for example, is cranial V (five).

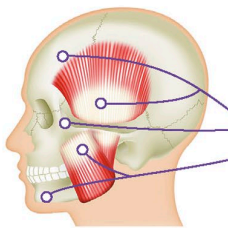
### Olfactory nerve (I, sensory)

Relays information about smells from the olfactory epithelium inside the nose, via the olfactory bulbs and tracts, to the brain's limbic centres.



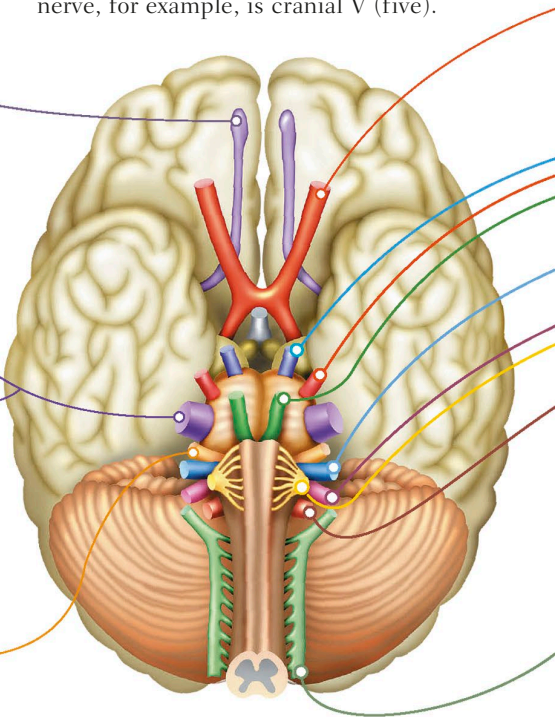
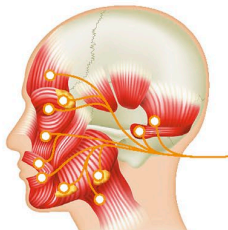
### Trigeminal nerve (V, two sensory and one mixed branch)

Ophthalmic and maxillary branches send signals from the eye, face, and teeth; mandibular fibres control chewing and send sensory signals from the lower jaw.



### Facial nerve (VII, mixed)

Sensory branches come from the taste buds of the front two-thirds of the tongue; motor fibres run to the muscles of facial expression and to the salivary and lacrimal glands.



## VIEW FROM BELOW

In this view of the underside of the brain, the pairs of cranial nerves are revealed as joining mainly to the lower regions of the brain. Some of these cranial nerves have a sensory function, taking impulses to

the brain. Others have a motor function, carrying nerve signals from the brain to various muscles and glands. Some cranial nerves are mixed, with both sensory and motor nerve fibres.